

Literature Status: Greedy-Envelope Counterexamples

Automatic Math Discovery Run `fa.banach.001`

June 12, 2026

Original Problems

The original source is Fernando Albiac, José L. Ansorena, Pablo M. Berná, and Przemysław Wojtaszczyk, *Greedy approximation for biorthogonal systems in quasi-Banach spaces*, arXiv:1903.11651; *Dissertationes Mathematicae* 560 (2021), 1–88.

In the arXiv PDF, the relevant questions appear as Problems 12.5 and 12.6. Problem 12.5 contains the general Banach-envelope question and a more special bidemocratic refinement:

sis. Given $0 < p < 1$, does there exist an infinite-dimensional subspace $\mathbb{X}_p \subset \mathbb{X}$ which has an equivalent p -norm?

Problem 12.5. Our discussion in Section 9 on Banach envelopes leaves the following open questions:

- (i) If $\mathcal{B}$ is a quasi-greedy basis in a quasi-Banach space $\mathbb{X}$, is its Banach envelope $\mathcal{B}_c$ quasi-greedy in the Banach envelope $\mathbb{X}_c$ of $\mathbb{X}$? More generally, is $\mathcal{B}_{c,r}$ quasi-greedy in the r -Banach envelope $\mathbb{X}_{c,r}$ for every (some) $0 < r \leq 1$?

The reader may argue that this question is too hard to tackle. Indeed, perhaps it would be more sensible to get started by focusing on special types of bases. In particular, the question remains open for bidemocratic quasi-greedy bases:

- (ii) Let $0 < r < 1$. If a basis is bidemocratic and quasi-greedy in $\mathbb{X}$, is it quasi-greedy in the r -Banach envelope $\mathbb{X}_{c,r}$?

Problem 12.6. The greedy algorithm with respect to bases is essentially independent of the particular ordering we chose for the basis. That is, an M -bounded semi-normalized basis $\mathcal{B} = (\mathbf{x}_n)_{n=1}^\infty$ enjoys the same greedy-like properties as any of its reorderings $(\mathbf{x}_{\phi(n)})_{n=1}^\infty$. Thus, when dealing with greedy-like properties, imposing conditions on the

Problem 12.6 contains the reordering question:

$\mathbb{A}$, is it quasi-greedy in the r -Banach envelope $\mathbb{A}_{C,r}$?

Problem 12.6. The greedy algorithm with respect to bases is essentially independent of the particular ordering we chose for the basis. That is, an M -bounded semi-normalized basis $\mathcal{B} = (\mathbf{x}_n)_{n=1}^{\infty}$ enjoys the same greedy-like properties as any of its reorderings $(\mathbf{x}_{\phi(n)})_{n=1}^{\infty}$. Thus, when dealing with greedy-like properties, imposing conditions on the basis such as being Schauder, which depend on a particular ordering, is somewhat unnatural. In fact, investigating the greedy algorithm without assuming that our basis is Schauder, not only does enable us to obtain more general results, but also, and above all, gives us the opportunity to differentiate those results that can be obtained only for Schauder bases from those other results that can be proved circumventing this condition. However, it must be conceded that the most important examples of bases respect to which the greedy algorithm is considered are Schauder bases, to the extent that the following question seems to be unsolved.

- Is there a quasi-greedy basis which is not a Schauder basis for any ordering?

Problem 12.7. Now we focus on locally convex quasi-Banach spaces. Every Schauder basis $\mathcal{B}$ for a Banach space is equivalent to its bidual basis $\mathcal{B}^{**}$ (see Theorem 1.14), but this result does not carry over to total M -bounded bases (see Proposition 1.11). This forced us to conduct our study of duality properties of greedy-like bases without this powerful

The first question asks, in its general part, whether a quasi-greedy basis in a quasi-Banach space remains quasi-greedy when passed to the Banach envelope. The second asks whether there can be a quasi-greedy basis which is not a Schauder basis for any ordering.

Supporting Answer Source

The supporting paper is Fernando Albiac, José L. Ansorena, Miguel Berasategui, and Pablo M. Berná, *When Greedy Approximation Breaks: Counterexamples in Quasi-Banach Spaces*, arXiv:2510.13693.

It cites the same questions using the published numbering as AABW Problems 13.5 and 13.6, and the AAT 2025 survey numbering as Problems 16 and 14:

regime, are naturally quasi-Banach, hence understanding the behaviour of greedy algorithms in these settings is therefore essential if the theory is to interact fully with harmonic analysis and approximation theory. Moreover, quasi-Banach spaces are intimately tied to their Banach envelopes, the closest Banach spaces into which they embed.

The relationship between a space and its envelope is often subtle and revealing, and the study of greedy approximation provides a new lens through which to explore this interaction. While in Banach spaces the theory is now mature, extending it to quasi-Banach spaces has revealed new phenomena and prompted a number of fundamental open problems. Two of these problems were posed in [1] and restated in the recent survey [4].

Question A ([1, Problem 13.5], cf. [4, Problem 16]). *Must a quasi-greedy basis in a quasi-Banach space remain quasi-greedy when considered in the Banach envelope?*

Question B ([1, Problem 13.6], cf. [4, Problem 14]). ([3, Problem 13.6], cf. [2, Problem 14]) *Can there exist a quasi-greedy basis that fails to be a Schauder basis under every possible reordering?*

In this paper we give counterexamples that resolve both problems. Our first construction shows that a quasi-Banach space $\mathbb{X}$ may carry an almost greedy Schauder basis whose image in the Banach envelope of $\mathbb{X}$ is not quasi-greedy. This demonstrates that the Banach envelope, while designed to “regularize” a quasi-Banach space, can significantly distort the behaviour of bases with respect to greedy approximation.

Answer

The questions checked in this packet are already answered in the negative by arXiv:2510.13693.

- Question A / AABW Problem 13.5 / AAT Problem 16 has a negative answer: there exists a quasi-Banach space X with an almost greedy Schauder basis whose image in the Banach envelope of X is not quasi-greedy.
- Question B / AABW Problem 13.6 / AAT Problem 14 has a positive existence answer: there exists a quasi-Banach space with an almost greedy basis which fails to be a Schauder basis under every rearrangement. Since an almost greedy basis is quasi-greedy, this answers the original quasi-greedy ordering question.

Verification Notes

The local source file

data/parsed/arxiv_sources/2510.13693/GreedyEnvelopeV02.tex

contains the following checks:

- lines 142–151 restate Question A and Question B and state that the paper gives counterexamples resolving both problems;
- the theorem collected near line 999 constructs the Banach-envelope counterexample and is followed by the sentence that it solves Question A in the negative;
- Theorem `thm:AMain`, around lines 1406–1416, constructs an almost greedy basis failing to be a Schauder basis under any rearrangement, thereby answering Question B in the negative.

Limitations

This packet is a literature-status packet, not an original proof or a new counterexample produced by this run. It also does not claim the bideocratic refinement inside the original Problem 12.5, since the supporting paper’s Question A is the general Banach-envelope problem. Finally, the supporting paper notes that the ordering question remains open when restricted to Banach spaces.

Search Bounds

The local run registry, solution packets, and attempts were searched for arXiv:2510.13693, the title phrase, the Banach-envelope quasi-greedy question, and Problems 13.5 and 13.6. No existing packet or registry row was found. Web search on June 12, 2026 located the arXiv records for arXiv:1903.11651, arXiv:2405.20939, and arXiv:2510.13693; the latter explicitly states in its abstract and introduction that it resolves the two earlier problems.

Human Review Recommendation

Reviewers should check the numbering mismatch between the arXiv preprint Problems 12.5–12.6 and the published AABW numbering cited by the answer paper as Problems 13.5–13.6. They should also verify that the packet is restricted to the general envelope question and the quasi-Banach ordering question, not to the bideocratic subproblem or the Banach-space-only variant.